MTU - Team “Delta L2”

AI-Based "Wingman" for Detecting PPE Violation and Emergency Situations on the Factory Floor

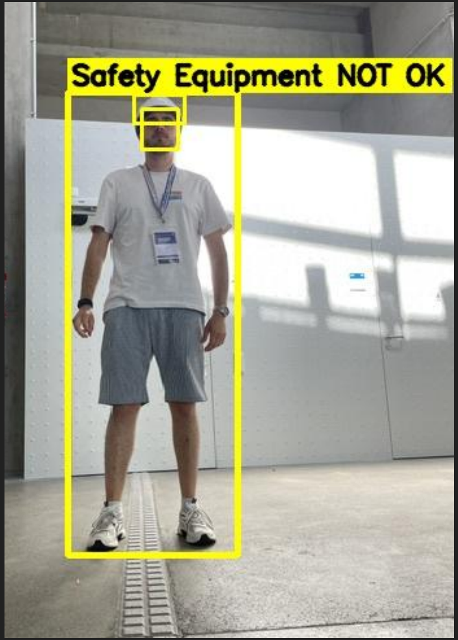
Problem

- Workers on factory floors may be required to wear personal protective equipment (PPE), such as hard hats and safety vests.
- Workers who fall or are incapacitated may go unnoticed for the first critical minutes of the incident

Our Solution

- AI based person detection and classification: Safety Equipment OK or NOK, Emergency
- Providing a dashboard as an intuitive overview
- Anonymization for compliance with the General Data Protection Regulation

Example



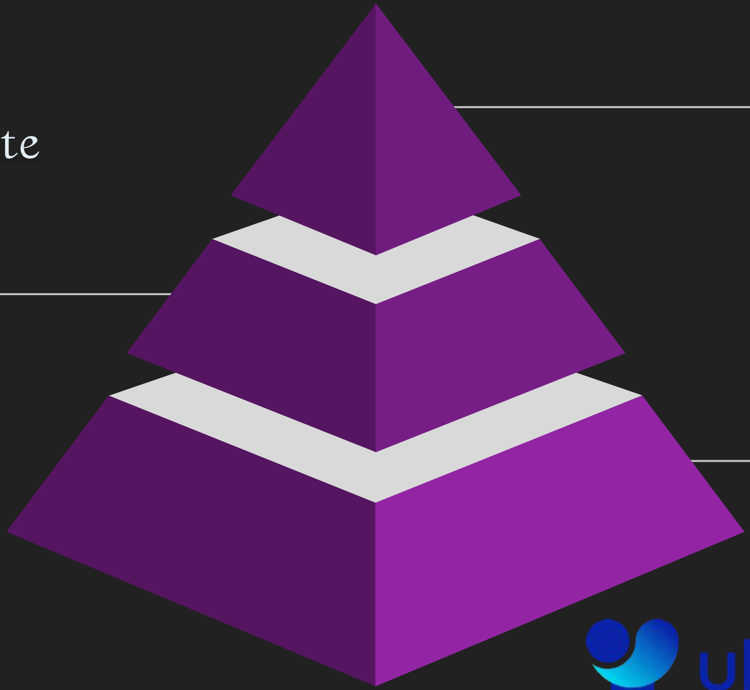
PPE Violation

Not wearing hardhat or not wearing a safety vest?

Emergency Situation

Is a Person lying on the Ground?

Architecture Overview



Frontend

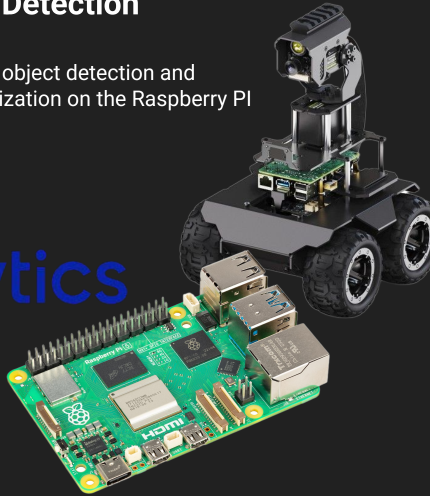
3 Real time monitoring of events, analytics for operator

Backend

2 Ingest events, manage Images and prepare data for presentation

Edge Detection

1 Custom object detection and anonymization on the Raspberry PI



Edge Detection

- Object Detection using custom YOLO26 models
- Optimized for problem specific classes
- More robust anonymization

Combined PPE Datasets

Detection of hardhats and safety vests

+

Face Detection Dataset

Model for face detection, more accurate for anonymization

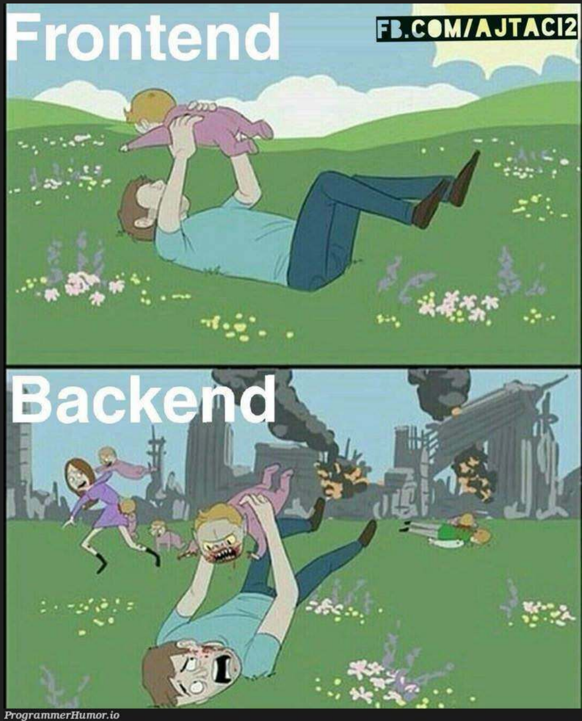
→

Wingman Model

Detects Person, Heads, Hard Hats, Vests with High accuracy

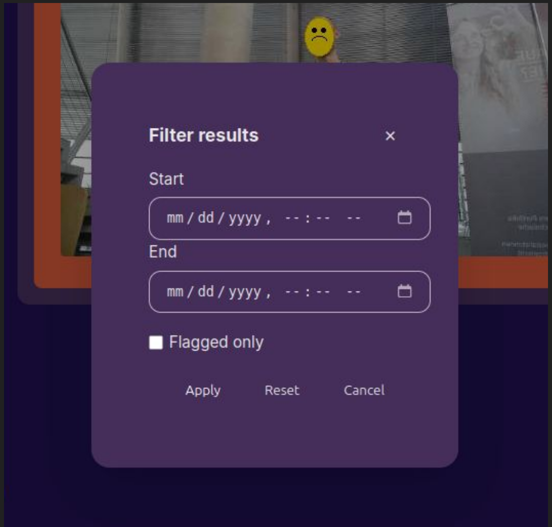
Backend

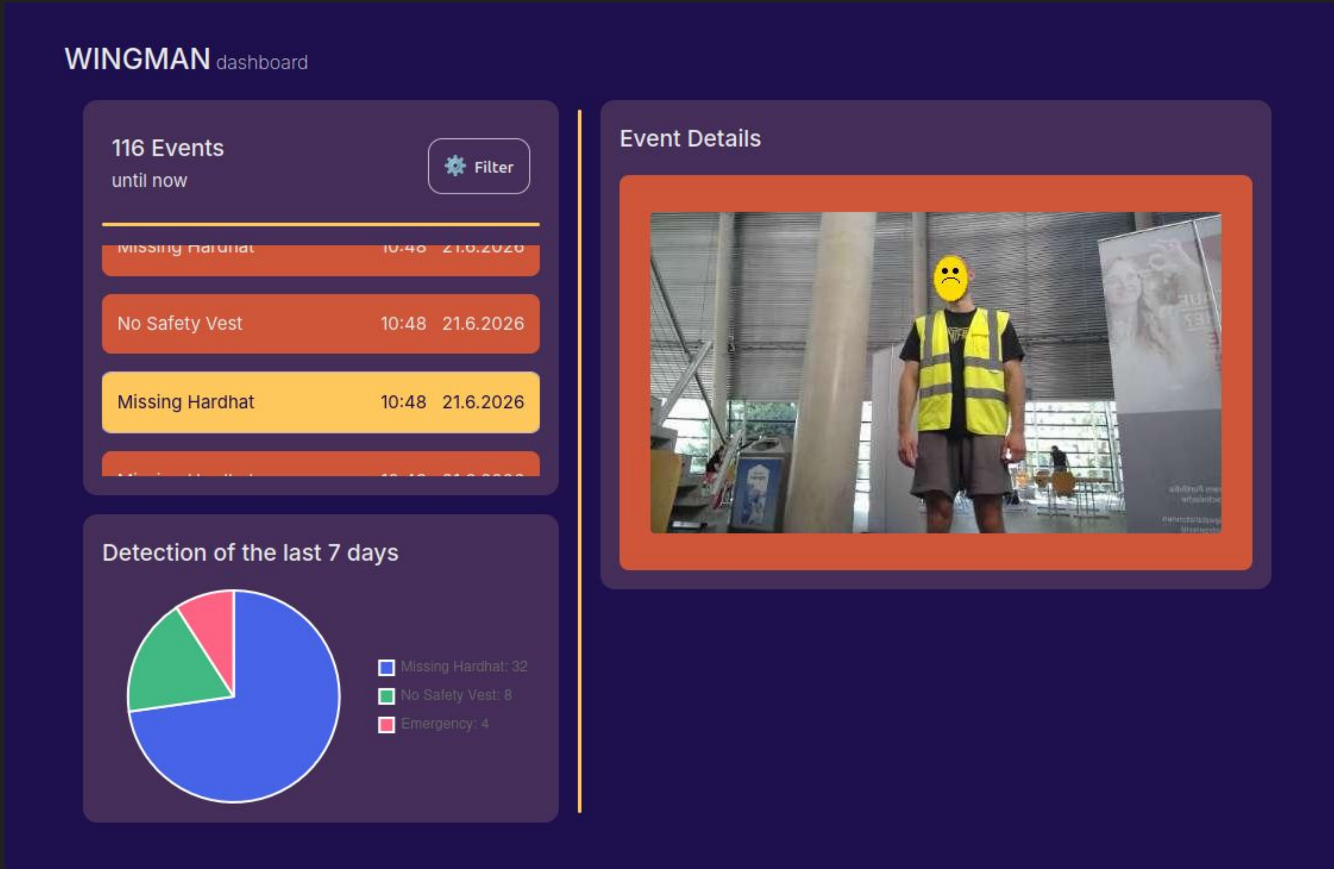
- Central processing of events based on FastAPI
- Stores events and anonymized images
- Prepares data for presentation
- REST-API



Dashboard

- Live Events
- Statistics
- Filtering





Demo

Summary

Self set baseline was achieved:

1. AI model on edge device
2. Backend receiving events and images
3. Frontend with the dashboard

… and there are a lot ideas for further development and improvement

→ Ask us about it 😉

Learnings - TOP 3

1. Priority Management → Only focus on the function and the pipeline
2. No overengineering → less custom code, more proven solutions
3. Steep learning curve with the necessity to get up to speed quickly

Thanks to Stefan and Martin for the supervision 🙌

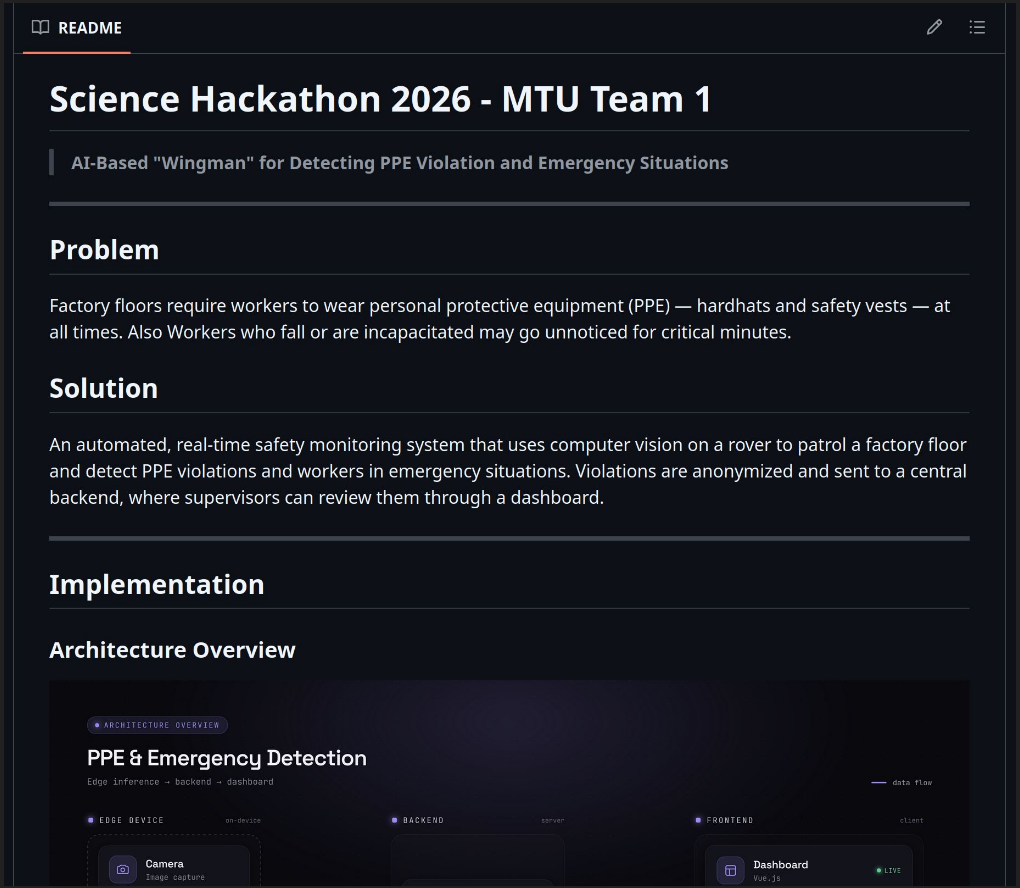
The Repository

Visit our repository on GitHub:

[mtuhack](https://github.com/jacobriege/mtuhack) - https://github.com/jacobriege/mtuhack

or

[HackAero/2026_mtu_team_1](https://github.com/HackAero/2026_mtu_team_1) - https://github.com/HackAero/2026_mtu_team_1


README

Science Hackathon 2026 - MTU Team 1

> **AI-Based "Wingman" for Detecting PPE Violation and Emergency Situations**

Problem

Factory floors require workers to wear personal protective equipment (PPE) — hardhats and safety vests — at all times. Also Workers who fall or are incapacitated may go unnoticed for critical minutes.

Solution

An automated, real-time safety monitoring system that uses computer vision on a rover to patrol a factory floor and detect PPE violations and workers in emergency situations. Violations are anonymized and sent to a central backend, where supervisors can review them through a dashboard.

Implementation

Architecture Overview